CROP RECOMMENDATION

USING MACHINE LEARNING

AgroSense AI • Agricultural Intelligence Platform

S. Abhinav • 2520080043

T. Harsha Vardhan • 2520080047

Guide: I v Sai Lakshmi Haritha, Assistant Professor, CS&IT

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College / University: [Add institution name]

Introduction

Precision agriculture + machine learning for data-driven crop selection

DOMAIN

Precision agriculture uses soil and environmental information to support cultivation decisions.

WHY NEEDED

Crop suitability depends on multiple parameters rather than a single soil or climate value.

PROJECT IDEA

Use a trained multiclass ML model to recommend a suitable crop from a 7-parameter field profile.

CORE INPUTS

N • P • K • Temperature • Humidity • Soil pH • Rainfall

Problem Statement

The system addresses crop-selection decisions that depend on interacting agricultural conditions.

EXISTING CHALLENGE

Multiple soil and climate parameters must be considered together.

A consistent computational pipeline is needed to turn these parameters into a crop recommendation.

LIMITATIONS / NEED

The recommendation must combine a trained ML model with a usable interface and contextual/live agricultural information.

Motivation

From static classification to an accessible agricultural decision-support platform

- Support informed cultivation decisions using measurable soil and environmental parameters.
- Provide a simple interface instead of requiring users to interpret a raw ML model.
- Combine the ML recommendation with geographic and live-climate context.
- Use Grok AI for contextual agricultural analysis and live recommendation support.
- Build a scalable full-stack platform rather than a standalone notebook.

Objectives

Project goals aligned with the supplied abstract and implemented platform

01 PREDICT

Recommend the most suitable crop from agricultural inputs.

02 PREPROCESS

Clean, normalize and select the seven model features.

03 EVALUATE

Measure accuracy, precision, recall and F1-score.

04 INTEGRATE

Connect React + Node.js with the ML recommendation workflow.

05 ENHANCE

Add geographic, climate and Grok AI contextual layers.

Existing System

Baseline approach reflected in the supplied abstract

BASELINE MODEL

Multinomial Logistic Regression (MLR) is used for supervised multiclass crop classification.

INPUTS

N, P, K, soil pH, temperature, humidity and rainfall.

LIMITATION OF A STANDALONE MODEL

A raw classifier does not by itself provide a complete user interface, geographic context, live climate display or AI-generated contextual explanation.

Our implementation turns the baseline ML idea into an integrated web platform.

Proposed System

AgroSense AI: a layered agricultural intelligence workflow

Layer 1	**Geographic Intelligence**	Region → State → District
Layer 2	**Live Climate**	Open-Meteo display layer
Layer 3	**ML Model**	7-feature soil/climate profile → crop
Layer 4	**Grok AI Insights**	Contextual agricultural analysis

System Architecture / Block Diagram

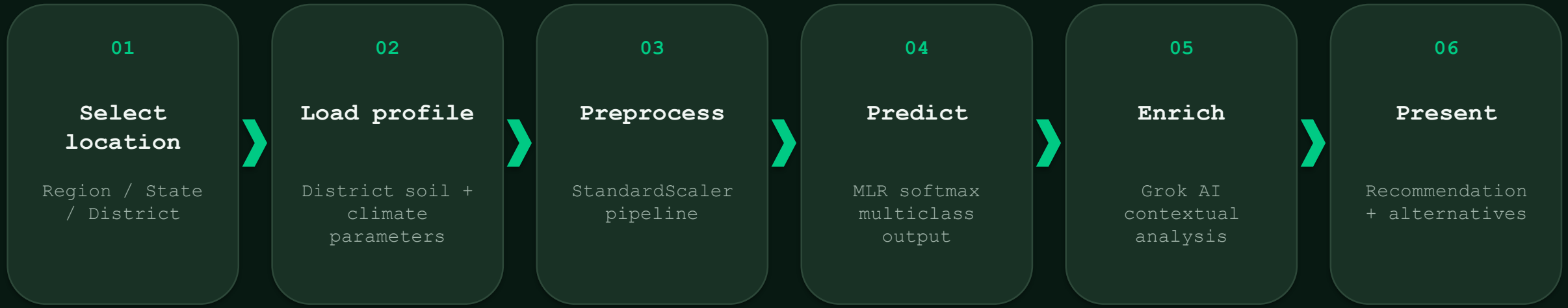
Input → full-stack processing → ML prediction → contextual output



Architecture shown in the project dashboard: geographic intelligence → live climate → ML model → Grok insights.

Methodology / Workflow

End-to-end project workflow



Technology Stack

Full-stack implementation + machine learning + AI integration

FRONTEND

React
Interactive dashboards, forms and recommendation UI

BACKEND

Node.js
API/orchestration layer connecting the web app to project services

ML

Python
Multinomial Logistic Regression
StandardScaler pipeline

AI + DATA

xAI Grok API for contextual agricultural insights
Crop Recommendation Dataset + district soil data
Open-Meteo for live climate display

System Modules / Components

Functional decomposition of the platform

User Interface

React-based farmer / meteorologist views

Geographic Intelligence

Region, state and district selection

Data Management

District soil profiles and dataset operations

Preprocessing

Standardization and feature pipeline

ML Prediction

Multinomial Logistic Regression

Decision Output

Crop, confidence and alternatives

Grok AI

Contextual agricultural analysis

Verification

Confusion matrix and model metrics dashboard

Implementation

Key implementation elements visible in the project platform

- React frontend provides the farmer and meteorologist interfaces.
- Node.js handles application/API orchestration.
- District soil data can be edited and saved through the meteorologist console.
- The ML pipeline uses StandardScaler followed by Logistic Regression.
- Model retraining is supported from the current dataset.
- The technical dashboard exposes confusion-matrix based verification.
- Grok AI is integrated for contextual agricultural insights.

LIVE PLATFORM

crop-ml-platform.vercel.app

AgroSense AI v4.0

Dataset / Input Data

Dataset details from the supplied abstract and deployed platform

DATASET

Crop Recommendation Dataset (Kaggle)
2,200 samples
22 balanced crop classes
100 records per crop

7 FEATURES

Nitrogen (N)
Phosphorus (P)
Potassium (K)
Temperature
Humidity
Soil pH
Rainfall

PREPROCESSING / SPLIT

Cleaning
Normalization / StandardScaler
Feature selection
80% train = 1,760
20% test = 440
Stratified by crop

Additional platform data: district-level soil profiles can be updated from the Meteorologist Console.

Algorithm / Model

Multinomial Logistic Regression with a standardized feature pipeline

PIPELINE

StandardScaler → LogisticRegression

MULTICLASS

Softmax probabilities across 22 crop classes.

PARAMETERS

Solver: lbfgs
max_iter: 1000

$$P(y=k) = e^{z_k} / \Sigma_j e^{z_j}$$

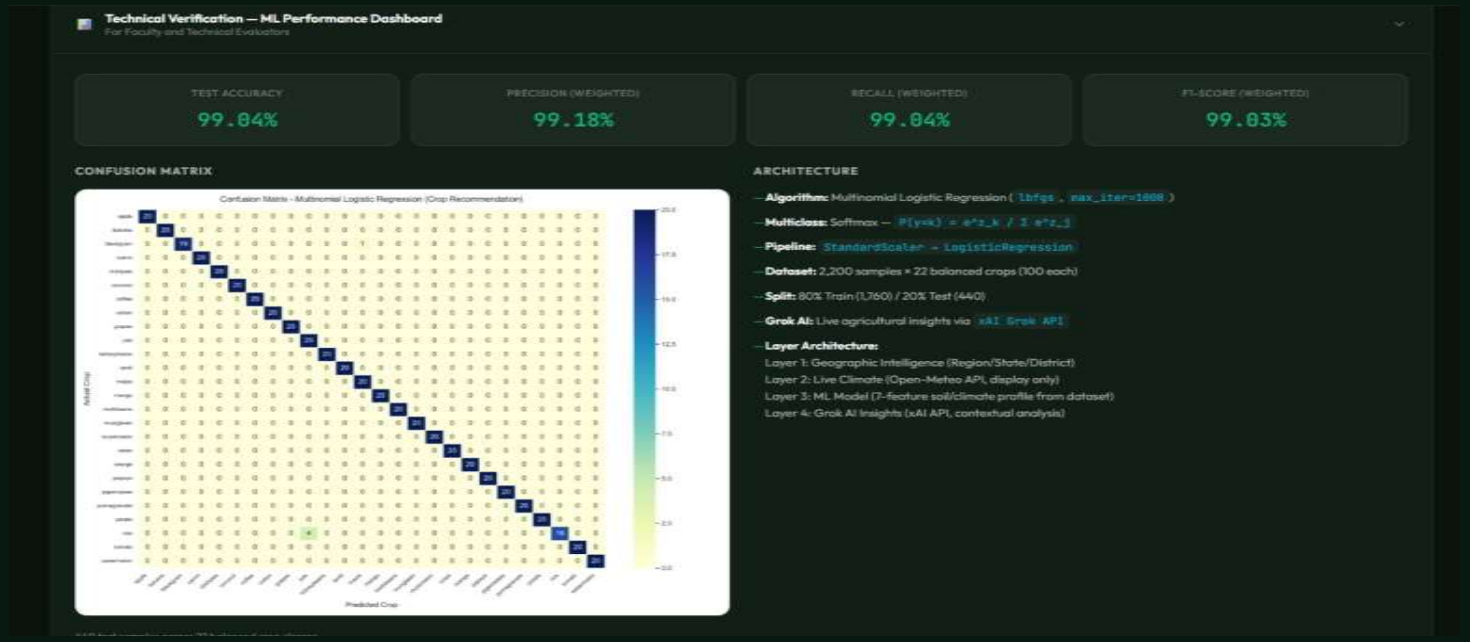
The model converts the seven-feature field fingerprint into a multiclass crop recommendation.

Results

Technical verification metrics shown in the supplied project dashboard screenshot

TEST ACCURACY	PRECISION (WEIGHTED)	RECALL (WEIGHTED)	F1-SCORE (WEIGHTED)
99.04%	99.18%	99.04%	99.03%

Test set: 440 samples across 22 balanced crop classes

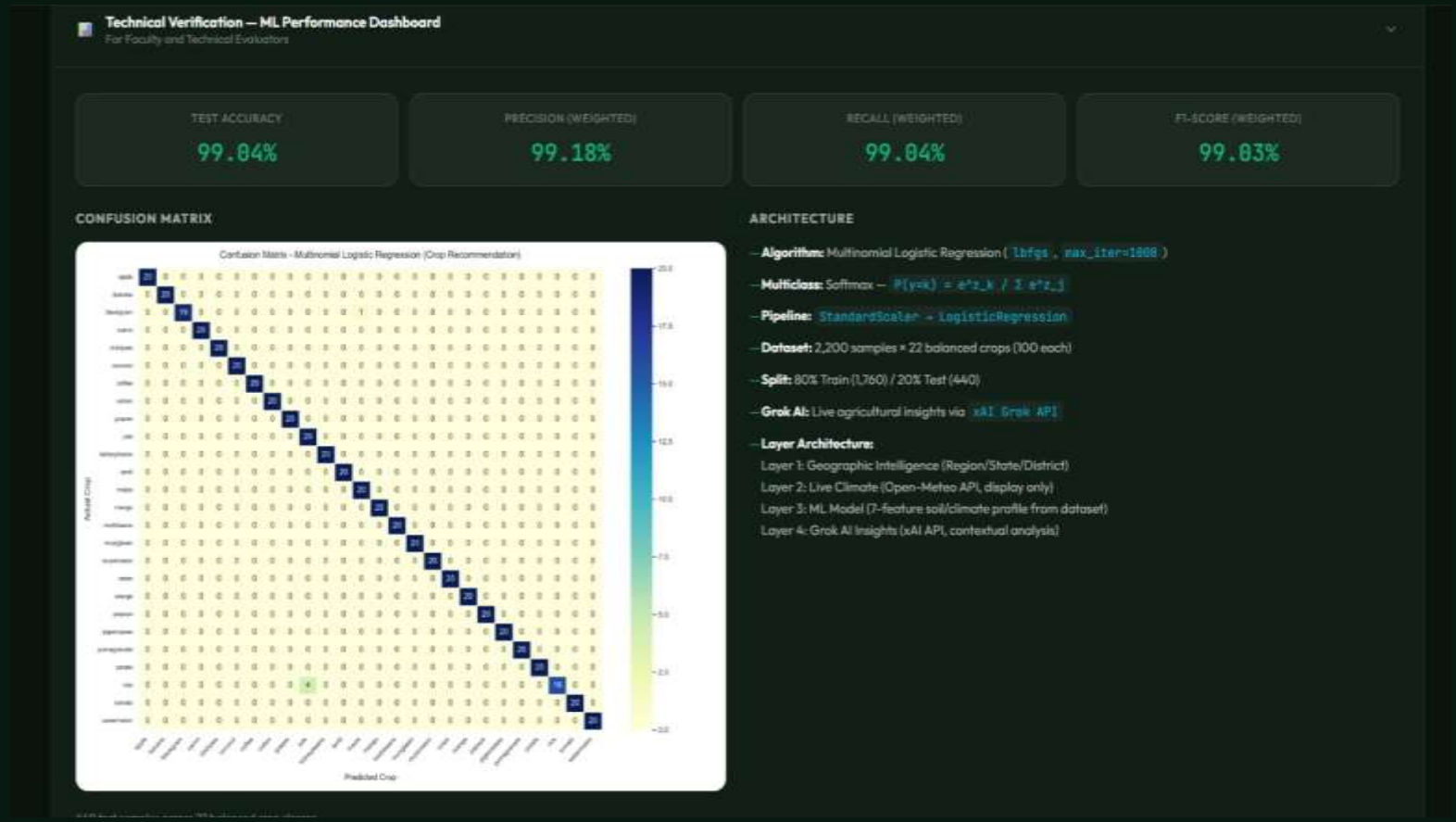


KEY OBSERVATION

The supplied dashboard shows a near-diagonal confusion matrix, indicating most test samples fall on their corresponding crop classes.

Screenshots / Demo

Project dashboard and deployed platform



DEPLOYED PLATFORM

AgroSense AI
https://crop-ml-platform.vercel.app/

Farmer view • Meteorologist Console • Technical Verification

Demo flow: Select location → load soil profile → get crop recommendation → view confidence/alternatives → inspect Grok AI analysis.

Advantages

Benefits of the integrated approach

- Combines a trained ML classifier with a practical web interface.
- Uses seven agricultural parameters as a combined field fingerprint.
- Provides crop confidence and alternative crop probabilities.
- Adds geographic context through region/state/district selection.
- Separates live climate display from the ML prediction layer.
- Adds Grok AI contextual analysis for richer agricultural guidance.
- Includes a technical verification dashboard for evaluators.

Applications

Potential real-world use cases of the platform

Farm Decision Support	Crop selection based on soil and environmental conditions.
Precision Agriculture	Use field profiles to support data-driven cultivation.
Agricultural Advisory	Combine model output with contextual AI insights.
Meteorologist / Analyst Console	Maintain district profiles and inspect model performance.
Educational / Demonstration	Showcase ML classification and full-stack integration.

Limitations

Current dataset, model and platform constraints

- The core model is trained on 22 crop classes from the supplied dataset.
- Recommendations are dependent on the quality and representativeness of available soil/climate data.
- Live climate data is architecturally separated from the ML model and is described as display-only in the platform.
- Grok AI provides contextual analysis; it should complement rather than replace the trained ML prediction.
- Real-world financial/economic values shown by the platform are indicative benchmarks, not guaranteed outcomes.
- Broader field validation across regions and seasons would be needed for production-scale agricultural decisions.

Future Scope

Possible improvements and scaling directions

More data

Expand crop classes, regions and seasonal observations.

Live fusion

Use live climate signals as an explicit model input after validation.

Model comparison

Benchmark MLR against additional supervised ML algorithms.

Mobile access

Extend the interface for field-first mobile workflows.

Explainability

Add stronger feature-level explanations for recommendations.

Deployment

Scale APIs, monitoring and model retraining workflows.

Conclusion

A full-stack crop recommendation platform built around a verified ML pipeline

WHAT WAS ACHIEVED

- Implemented crop recommendation using Multinomial Logistic Regression.
- Used seven soil/environmental features across 22 balanced crop classes.
- Built an integrated React + Node.js application.
- Added geographic intelligence, live climate display and Grok AI contextual analysis.
- Provided a technical verification dashboard with model metrics and confusion matrix.

OVERALL CONTRIBUTION

A practical, scalable decision-support platform that connects machine learning, agricultural data, a modern web interface and AI-assisted contextual guidance.

Thank You

Questions & Discussion